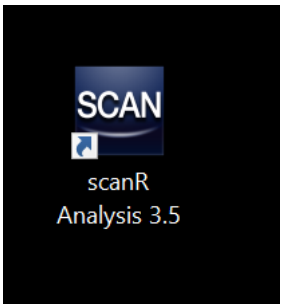
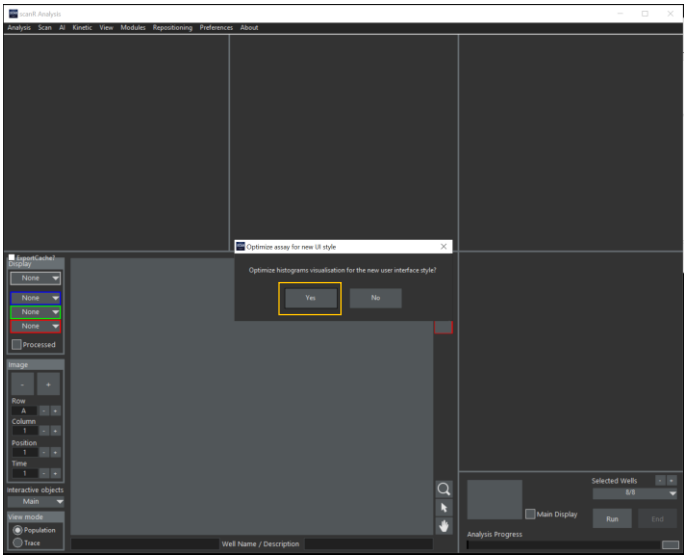
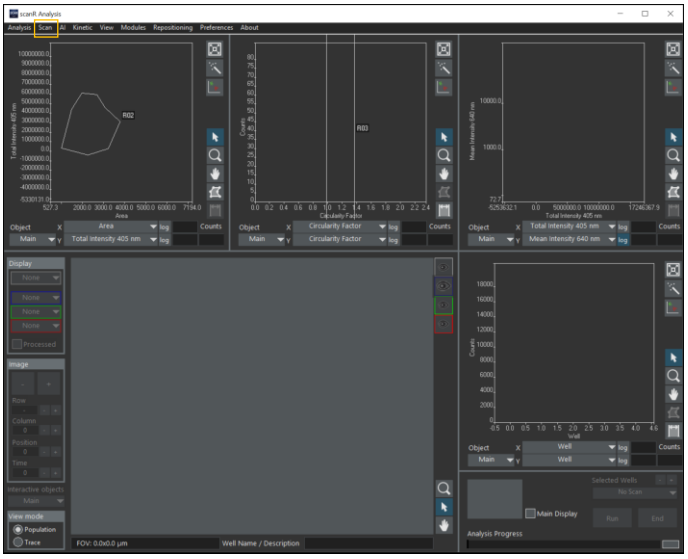
Šárka
Janscak's group

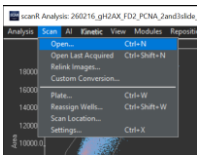


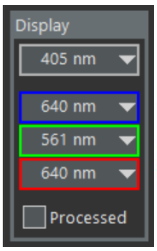


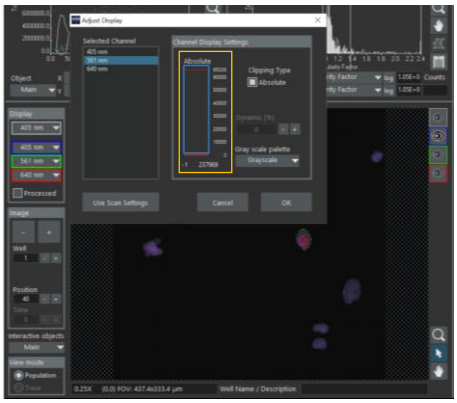


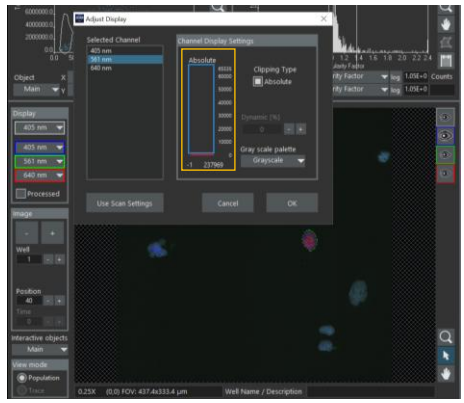
Load data from microscope
Scan → Open … (Ctrl + N) → find Folder → experiment_descriptor.xml

Or open previous analysis (to apply previous settings)
Analysis → Open









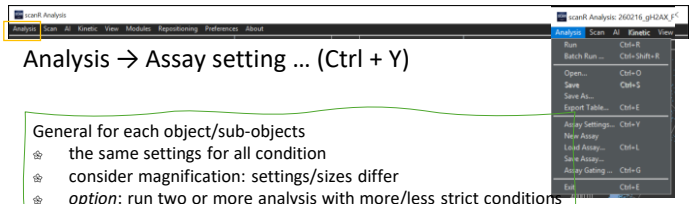
Display

- colors can be combined e.g. blue + red = magenta

View → Adjust Display

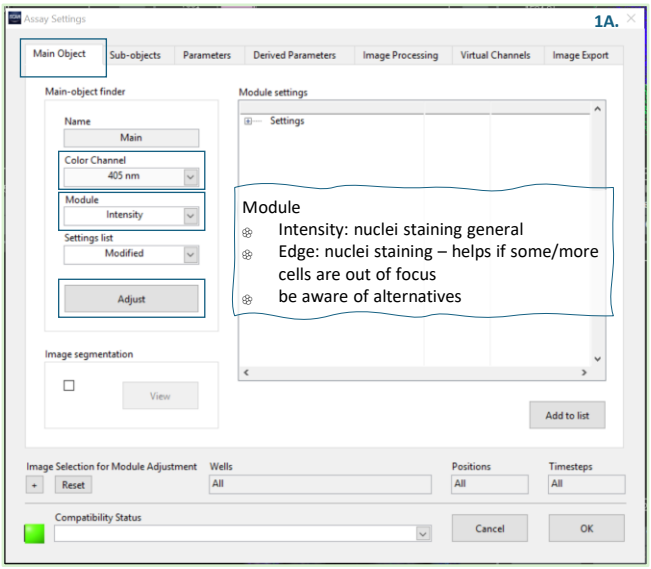
- Adjust intensity (usually it is necessary to go very low)
- it has to be done for each channel separately

- can be done also later (for gating)

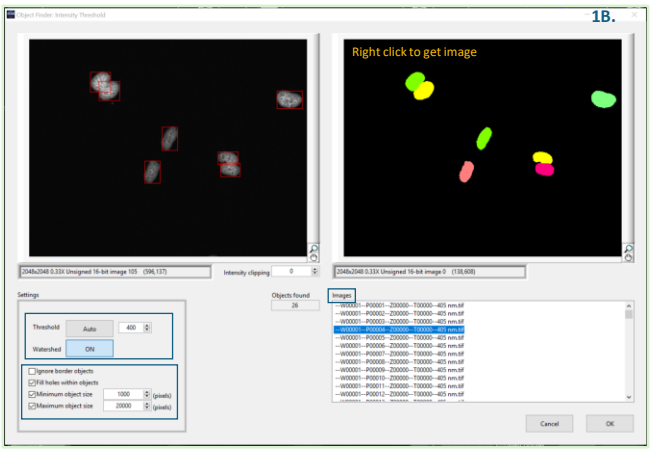


Analysis → Assay setting … (Ctrl + Y)

General for each object/sub-objects
- the same settings for all condition
- consider magnification: settings/sizes differ
- *option*: run two or more analysis with more/less strict conditions



Module
- Intensity: nuclei staining general
- Edge: nuclei staining – helps if some/more cells are out of focus
- be aware of alternatives



Threshold
- intensity o signal against background

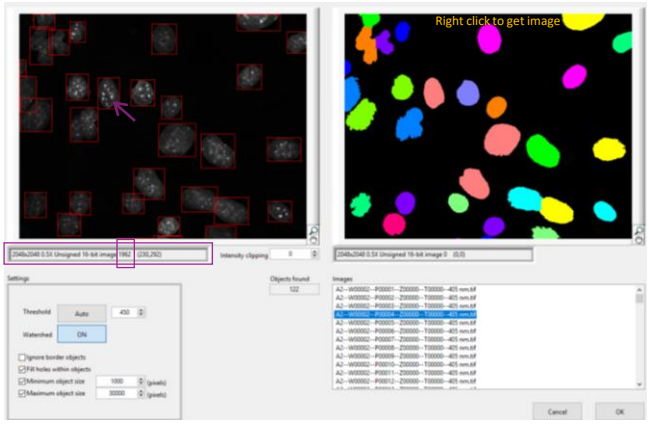
Ignore borders
- optional, can be gated out

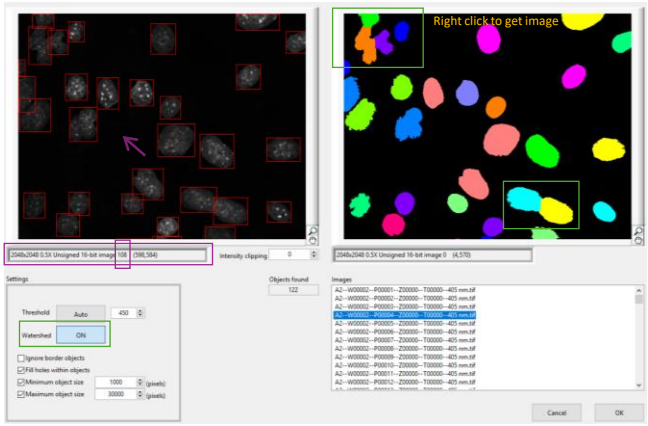
Watershed ON
- to separate object close or together

Minimum and Maximum object size
- to find almost all cells – it can be excluded later in gating

Images
W0000X … X … individual well/cover slip
P0000X … image
- go thought a few images and different treatments/ctrl to see, if settings are fine for all conditions

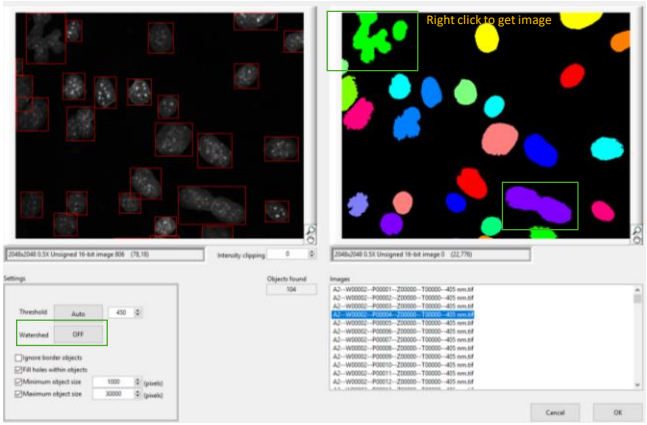
Example:

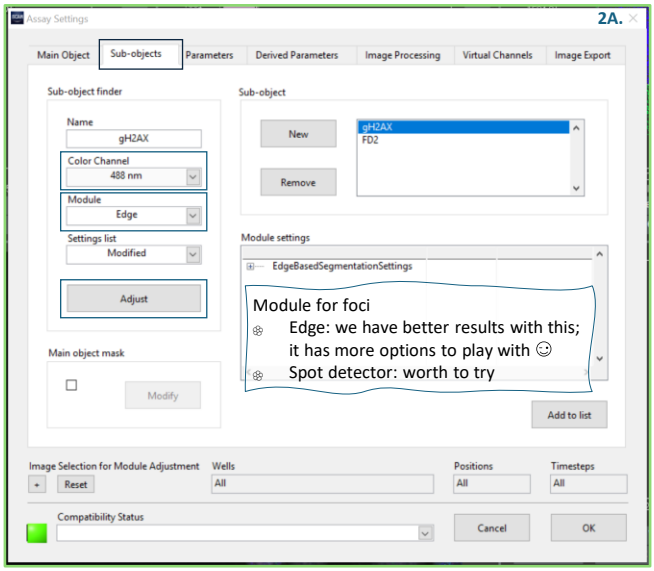


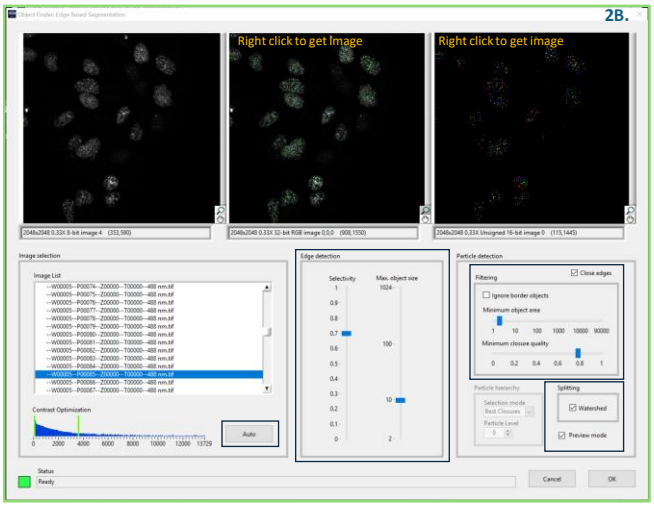


Threshold
- intensity of signal objects against background
- point with cursor on the object/background to see intensity
- ideally difference approx. 6-times

Watershed ON
- to separate object close or together







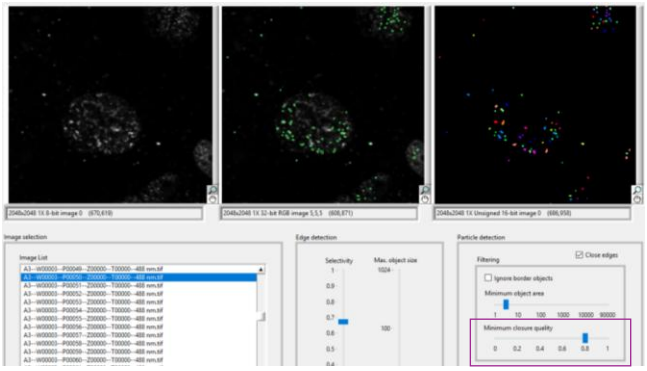
Auto

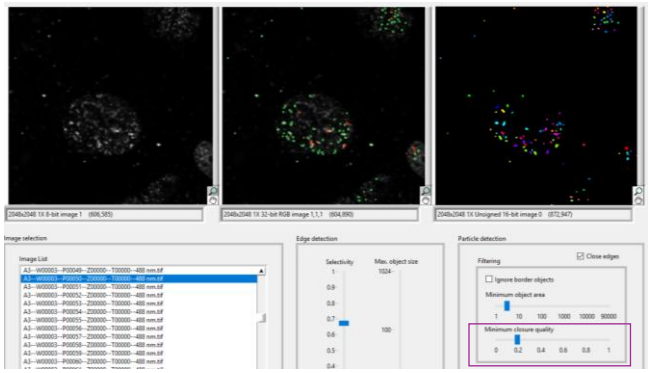
- set it to see, but also adjust brightness manually – same range for all condition

Edge & particle detection

- Adjust based on desired objects
- tip : zoom in: to be sure, which foci are masked

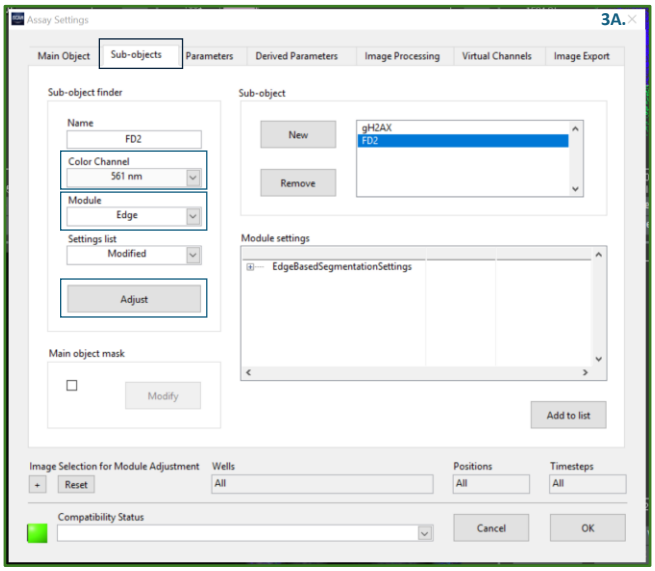
Example:

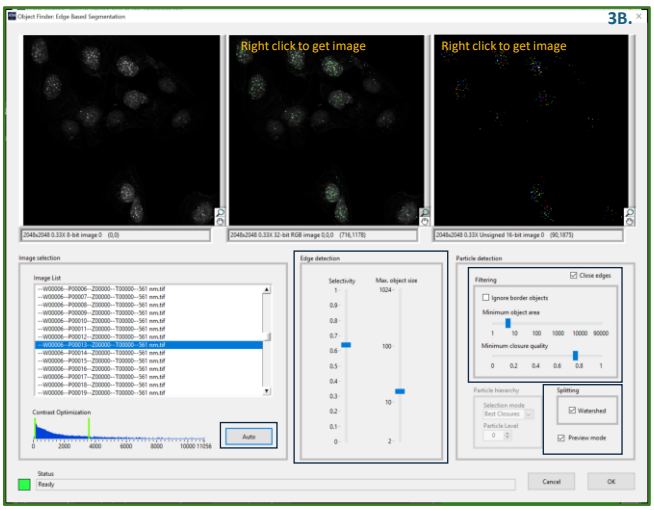




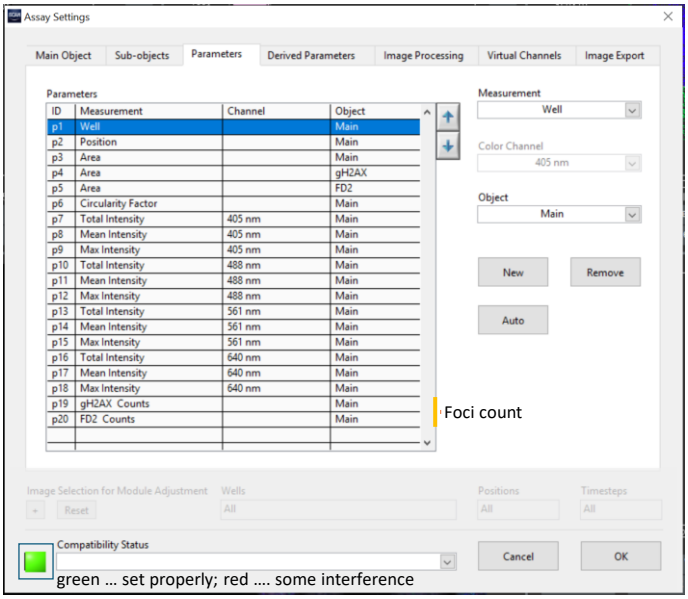
Minimum closure quality

- lower → pick more "objects", but can detect undesirable
- be aware, what software is picking across images





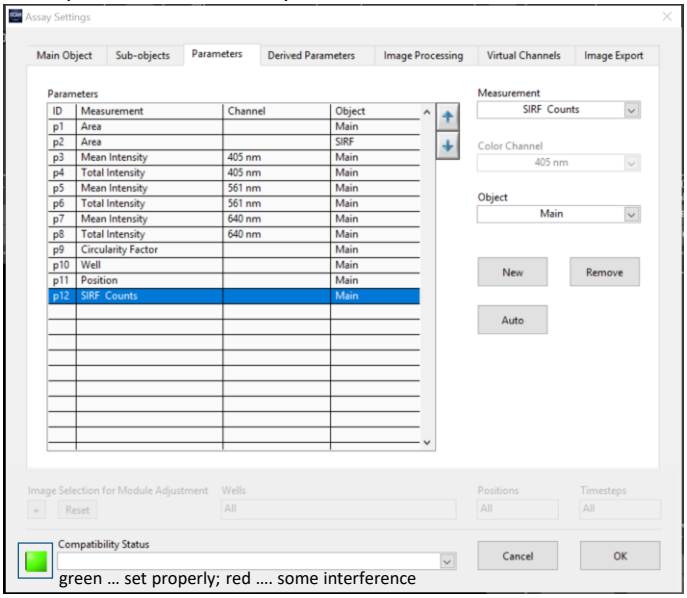
Foci



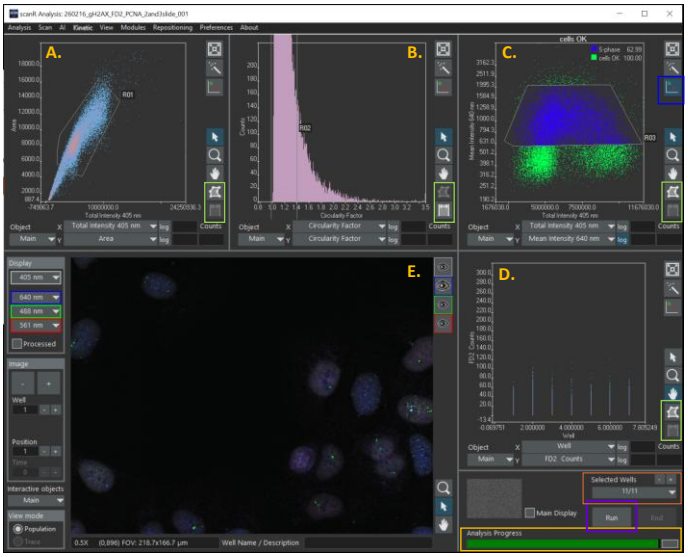
Foci count

green … set properly; red …. some interference

Example of PLA/SIRF assay



green … set properly; red …. some interference



once Parameters set → OK → **Run** analysis

Change color based on gate(s)

Gates drawing

- double click by mouse to close gate ("circular" ones)

Gating

A: selection of cells

- exclude debris (down) or doublets (up) etc.

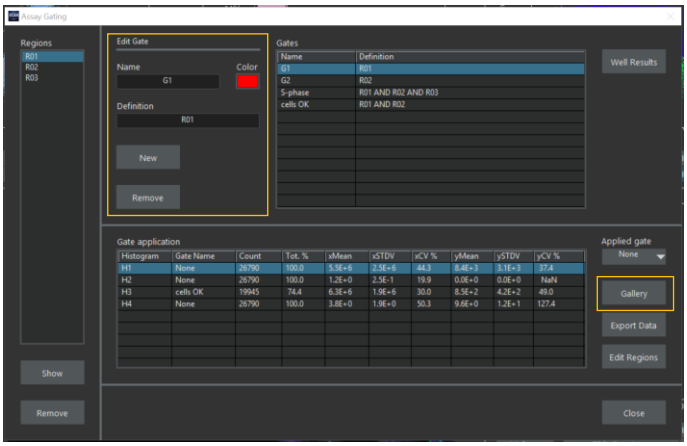
B. selection of circular cells (base on circularity factor)

C. Cell cycle (X: Total intensity DAPI / Y: Mean intensity PCNA/EdU)

D. Results: condition (Y axis) / well (X axis)

- Selected wells – enables to see individual conditions
- clicking on the individual points in each plots show the image of object in the view window (***E***)

Analysis → Assay Gating …

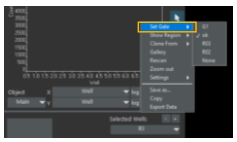


Applying gating

Analysis → Assay Gating …

- define gates; it has to be written properly (capital letters)
- apply gating: right click on plot (A – C) → select desired gate

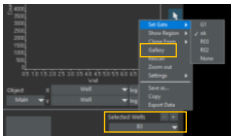
Export Gallery (option I)

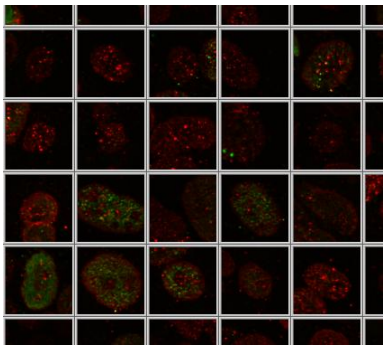


Export Gallery (option II)

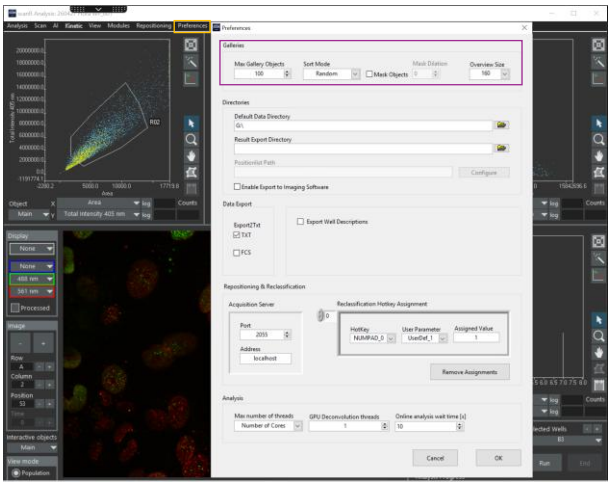
- according active window/field

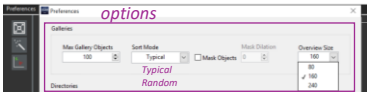
option: right click on well in plot D.





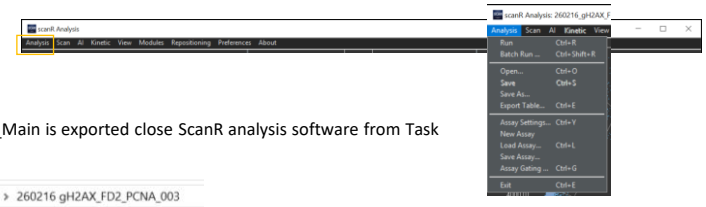
Setting for exporting galleries





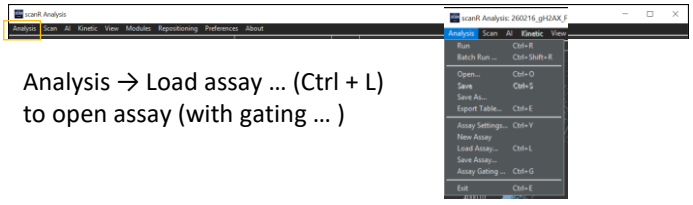
Finish analysis

- Save as… /Save (Ctrl+S): .sca
- Save Assay… : .say
- Export Table… (Ctrl+E): _Main.txt
  - before export (.txt) be sure everything is saved !!!!
  - it will export all results in separate .txt files, once _Main is exported close ScanR analysis software from Task manager (it can take long time)

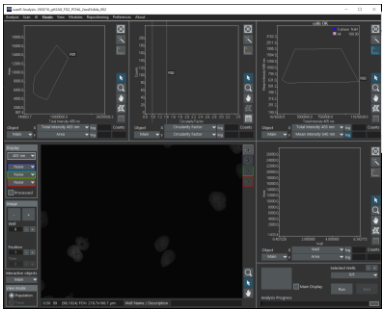


Name	Date modified	Type	Size
data	16.02.2026 15:46	File folder	
260216 gH2AX_FD2_PCNA_003.say	07.04.2026 13:00	SAY File	61 KB
260216 gH2AX_FD2_PCNA_003.sca	07.04.2026 12:57	SCA File	67'865 KB
260216 gH2AX_FD2_PCNA_003_gH2AX.txt	07.04.2026 13:06	Text Document	14'039 KB
260216 gH2AX_FD2_PCNA_003_Main.txt	07.04.2026 13:01	Text Document	3'404 KB
AcquisitionLog.dat	16.02.2026 15:46	DAT File	49 KB
experiment_descriptor.dat	16.02.2026 15:05	DAT File	39 KB
experiment_descriptor.xml	16.02.2026 15:05	XML Document	69 KB

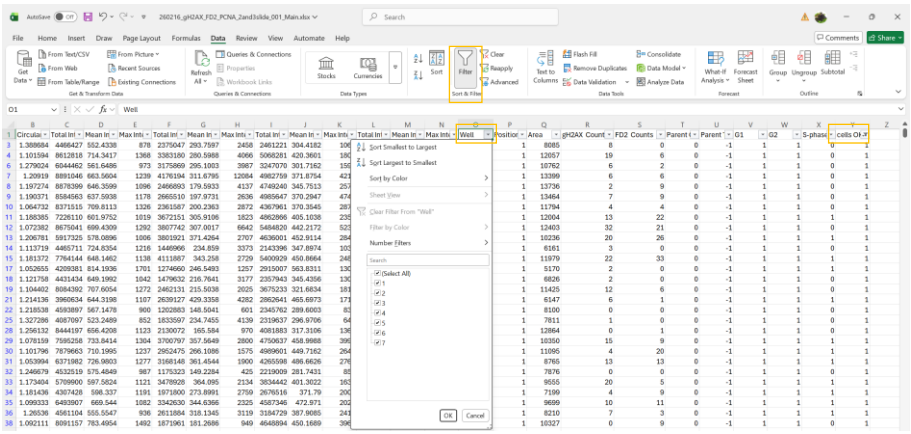
Note: Loading saved assay (.say)



Analysis → Load assay … (Ctrl + L)
to open assay (with gating …)



Note: Example of further data processing (.txt)



Open _Main.txt file in Excel

Apply filters
Well … condition

Tip: Freeze top row (view)

Copy individual values for each condition to a new sheet

Filter
general: well and cells OK (from gating)
can be another ….

	A	B	C	D	E	F	G	H	I
1	260216 gH2AX_FD2_PCNA_003								
2	NT		CPT		CDDP		MMC		
3	well 1		well 2		well 3		well 5		
4									
5									
6		17.59587		52.93338		23.65166		43.20109	
7		3677		5284		4042		3655	
8		647		2797		956		1579	
9	7.994017	16.48001	43.71631	33.64269	30.52251	19.30529	39.67469	27.43256	
10	H2AX foci	FD2 foci C	H2AX foci	FD2 foci C	H2AX foci	FD2 foci C	H2AX foci	FD2 foci Counts	
11	15	26	35	28	29	14	79	38	
12	26	21	46	69	6	17	41	45	
13	9	20	0	1	7	11	26	26	

Example
count FD2 foci >30

=D8/D7*100
=COUNT(D11:D140002)
=COUNTIF(D11:D19002,">30")

Important !!!
Save as excel file, otherwise _Main.txt will be overwritten

	Object	Mean In	Total In	Mean In	Total In	Mean In	Total In	Circulai	Area	Well	Positior	H2AX C	Parent	Parent	G1	ok
15800	15798	740.9567	2905291	1215.821	4767235	144.4019	566200	1.118277	3921	7	1	0	6	-1	1	1
15803	15801	839.6082	9103871	961.2178	10422484	994.3918	10782190	1.221777	10843	7	1	5	6	-1	1	1
15807	15805	1064.25	4781676	1318.108	5922260	826.4747	3713351	1.176879	4493	7	1	15	6	-1	1	1
15808	15806	826.8613	2909725	1336.812	4704241	238.8897	840653	1.287088	3519	7	1	1	6	-1	1	1

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1		1				2				3				4			
2																	
3		A2				A3				B2				B3			
4																	
5	% of cells with more than 10 H2			6.017358				9.503855				19.69803				15.59083	
6	No of cells > 10 foci			312				567				561				442	
7	No of cells			5185				5966				2848				2835	
8	Average	434.8112		2.207715		604.4163		3.456755		1048.938		6.950492		771.0257		5.449383	
9																	
10		Mean H2A	Total H2A	H2AX foci		Mean H2A	Total H2A	H2AX foci		Mean H2A	Total H2A	H2AX foci		Mean H2A	Total H2A	H2AX foci	
11		191.5316	1835256	0		781.6293	2865453	2		219.141	2490756	0		144.4019	566200	0	
12		374.1531	3231560	2		240.2176	2051458	0		1768.509	16977682	24		994.3918	10782190	5	
13		211.0323	568521	0		553.8734	6301417	0		542.2919	2799311	1		826.4747	3713351	15	
14		180.4937	1816489	2		560.2778	5426851	0		186.8094	1370434	0		238.8897	840653	1	

=D6/D7*100
=COUNTIF(D11:D15000,">10")
=COUNT(D11:D14000)
=AVERAGE(D11:D14000)